FINANCIAL MANAGEMENT

Suggested Answers
November-December 2023

Answer to the Question# 1(a) (i):

Calculating the Cost of equity	
Rf	9%
Market Risk premium	3%
Beta	1.2
Additional premium for Local investments	3%
Ke (For local investments)	**15.60%**

Answer to the Question# 1(a) (ii):

Investment in Father's friend business							
							In BDT Millions
	Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**
Investment	-3,000						
Return		450	450	450	450	450	450
At maturity							3,000
Additional Return							210
NCF	**-3,000**	**450**	**450**	**450**	**450**	**450**	**3,660**
DCF at 15.6% (W-8)	1	0.865	0.748	0.647	0.56	0.484	0.419
PV	-3,000	389	337	291	252	218	1,534
NPV	**20**						

Answer to the Question# 1(b):

Basic data & preliminary workings									
		Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Royalty payable	USD	-	-	25,000	25,000	25,000	25,000	25,000	
Exchange rate	SAR/US	3.75							
	BDT/US	110							

Working Capital - (In Millions) (SAR)									
Working capital			27.5	11.96	12.44	12.94	13.45	13.99	
Incremental working capital				-15.54	0.48	0.5	0.51	0.54	13.99

Depreciation (In Millions) (SAR)									
Machinery			20.000	16.000	12.800	10.240	8.192	6.554	5.243
Depreciation on reducing balance method			4.000	3.200	2.560	2.048	1.638	1.311	1.049
Tax savings 20%				0.80	0.64	0.51	0.41	0.33	0.26

Purchase consideration (SAR)									
KSA Exchange rate		3.75	3.713	3.679	3.645	3.608	3.57	3.54	
Purchase consideration (6.5 mn*3.54) (SAR)								**23.01**	**Million**

Royalty Income for REBAN									
Year		**Y0**	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Royalty received (USD) in Bangladesh				25,000	25,000	25,000	25,000	25,000	
Inflation- US	4%	1	1.040	1.082	1.125	1.170	1.217	1.265	1.316
Inflation- Bangladesh	9%	1	1.090	1.188	1.295	1.412	1.539	1.677	1.828
			1.048	1.098	1.151	1.207	1.265	1.326	1.389

Ex.rate & Inflation rate adjustment BDT/USD		110	115.28	120.78	126.61	132.77	139.15	145.86	152.79
Royalty income (BDT Millions)		-	-	**3.020**	**3.165**	**3.319**	**3.479**	**3.647**	-

Royalty Cost for REKSA									
Inflation rate USA	4%	1	1.040	1.082	1.125	1.170	1.217	1.265	1.316
Inflation rate KSA	3%	1	1.030	1.061	1.093	1.126	1.159	1.194	1.230
			0.990	0.981	0.972	0.962	0.952	0.944	0.935
Ex.rate & Inflation rate adjustment SAR/USD		3.75	3.713	3.679	3.645	3.608	3.57	3.54	3.506
Royalty cost (In Millions) (SAR)				**0.092**	**0.091**	**0.09**	**0.089**	**0.089**	**0**

Answer to the Question# 1(c) (i):

Working 08: Calculating the Cost of equity	
Rf	9%
Market Risk premium	3%
Beta	1.2
Additional premium for Local investments	3%
Ke (For local investments)	**15.60%**
Foreign investments	2%
Ke (For foreign investments)	**17.60%**

Answer to the Question# 1(c) (ii):

KSA Operations								
	Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Units	-	-	15,000	21,000	26,000	26,000	26,000	
Conribution per unit 2700 (SAR)			2,700	2,700	2,700	2,700	2,700	
Inflation at 3% p.a	1	1.03	1.061	1.093	1.126	1.159	1.194	1.23
	In Millions SAR							
Total Contribution	-	-	**42.97**	**61.97**	**79.05**	**81.36**	**83.82**	-
Adm,S&D 7.5%of Tot.Cont			3.22	4.65	5.93	6.10	6.29	
Royalty W-1			0.09	0.09	0.09	0.09	0.09	
Depreciation W7			4.00	3.20	2.56	2.05	1.64	
Total Cost (SAR)			**7.31**	**7.94**	**8.58**	**8.24**	**8.01**	-
EBIT			35.66	54.03	70.47	73.12	75.80	-
Income tax @ 20% (Working 04)				(7.93)	(11.45)	(14.61)	(15.03)	(15.49)
Profit After Tax (accounting profit)			**35.66**	**46.10**	**59.02**	**58.52**	**60.77**	**(15.49)**
Cash profit after tax (Working 04)			**39.66**	**49.30**	**61.58**	**60.57**	**62.41**	**(15.49)**

Operations in KSA (In Millions)									
Contribution		-	-	42.97	61.97	79.05	81.36	83.82	-
Administration, selling & distribution				(3.22)	(4.65)	(5.93)	(6.10)	(6.29)	
Royalty cost				(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	
Cash profit before tax				39.66	57.23	73.03	75.17	77.44	-
Income tax at 20%					(7.93)	(11.45)	(14.61)	(15.03)	(15.49)
Cash profit after tax (SAR)				**39.66**	**49.30**	**61.58**	**60.57**	**62.41**	**(15.49)**

Answer to the Question# 1(d):

Cash Outflow: Investments								
Land	-10							
Buildings	-7.5	-15						
Machinery		-20						
Working Capital (Working 06)		-27.5	15.54	-0.48	-0.5	-0.51	-0.54	13.99
Realizable Value (Purchase consideration) (Working 05)							**23.01**	
Tax benefit from Machinery (Working 07)			0.80	0.64	0.51	0.41	0.33	0.26
Net Cash Flow (in SAR)	**-17.5**	**-62.5**	**56.00**	**49.46**	**61.59**	**60.46**	**85.21**	**(1.24)**
Exchange Rate BDT/SAR W9	29.333	31.048	32.830	34.735	36.799	38.978	41.203	43.580
Net Cash Flows remitted to Bangladesh	**(513.33)**	**(1,940.48)**	**1,838.32**	**1,718.11**	**2,266.51**	**2,356.78**	**3,510.81**	**(53.89)**

								In BDT Million
	Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Royalty income	-	-	3.020	3.165	3.319	3.479	3.647	-
Operational	-	-	(1.188)	(1.943)	(2.683)	(2.924)	(3.186)	-
Net Cash flow	-	-	**1.832**	**1.223**	**0.636**	**0.555**	**0.460**	-
Tax expense (22.5%)			-	0.412	0.275	0.143	0.125	0.104
Net Cash flows after			**1.832**	**0.811**	**0.361**	**0.411**	**0.335**	**(0.104)**
Cash remitted (from)/to Bangladesh	**(513.33)**	**(1,940.48)**	**1,838.32**	**1,718.11**	**2,266.51**	**2,356.78**	**3,510.81**	**(53.89)**
Total cash flows	(513.33)	(1,940.48)	1,840.15	1,718.92	2,266.87	2,357.19	3,511.15	(53.99)
DCF at 17.6%	1	0.850	0.723	0.615	0.523	0.445	0.378	0.321
PV	(513.33)	(1,649.41)	1,330.43	1,057.14	1,185.57	1,048.95	1,327.21	(17.33)
NPV	**3,769**							

WORKINGS:

Basic data & preliminary workings									
		Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Royalty payable	USD	-	-	25,000	25,000	25,000	25,000	25,000	
Exchange rate	SAR/US	3.75							
	BDT/US	110							

Working 01	Royalty Cost for REKSA								
Inflation rate USA	4%	1	1.040	1.082	1.125	1.170	1.217	1.265	1.316
Inflation rate KSA	3%	1	1.030	1.061	1.093	1.126	1.159	1.194	1.230
			0.990	0.981	0.972	0.962	0.952	0.944	0.935
Ex.rate & Inflation rate adjustment SAR/USD		3.75	3.713	3.679	3.645	3.608	3.57	3.54	3.506
Royalty cost (In Millions) (SAR)				**0.092**	**0.091**	**0.09**	**0.089**	**0.089**	**0**

Working 02	Royalty Income for REBAN								
Year		**Y0**	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**
Royalty received (USD) in Bangladesh				25,000	25,000	25,000	25,000	25,000	
Inflation- US	4%	1	1.040	1.082	1.125	1.170	1.217	1.265	1.316
Inflation- Bangladesh	9%	1	1.090	1.188	1.295	1.412	1.539	1.677	1.828
			1.048	1.098	1.151	1.207	1.265	1.326	1.389
Ex.rate & Inflation rate adjustment BDT/USD		110	115.28	120.78	126.61	132.77	139.15	145.86	152.79
Royalty income (BDT Millions)		-	-	**3.020**	**3.165**	**3.319**	**3.479**	**3.647**	**-**

Working 03	Administrative & other expenses of Bangladesh operation								
Inflation effect (9%)		1	1.09	1.188	1.295	1.412	1.539	1.677	
Administrative expenses at current prices (In Millions)		-	-	1.00	1.5	1.9	1.9	1.9	1.9
Administrative expenses (In BDT Millions)		-	-	**1.188**	**1.943**	**2.683**	**2.924**	**3.186**	-

Working 04 - (In Millions)	Operations in KSA								
Contribution		-	-	42.97	61.97	79.05	81.36	83.82	-
Administration, selling & distribution				(3.22)	(4.65)	(5.93)	(6.10)	(6.29)	
Royalty cost				(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	
Cash profit before tax				39.66	57.23	73.03	75.17	77.44	-
Income tax at 20%					(7.93)	(11.45)	(14.61)	(15.03)	(15.49)
Cash profit after tax (SAR)				**39.66**	**49.30**	**61.58**	**60.57**	**62.41**	**(15.49)**

Working 05- Purchase consideration (SAR)									
KSA Exchange rate		3.75	3.713	3.679	3.645	3.608	3.57	3.54	
Purchase consideration (6.5 mn*3.54) (SAR)								**23.01**	**Million**

Working 06- Working Capital - (In Millions) (SAR)									
Working capital			27.5	11.96	12.44	12.94	13.45	13.99	
Incremental working capital				-15.54	0.48	0.5	0.51	0.54	13.99

Working 07- Depreciation (In Millions) (SAR)									
Machinery			20.000	16.000	12.800	10.240	8.192	6.554	5.243
Depreciation on reducing balance method			4.000	3.200	2.560	2.048	1.638	1.311	1.049
Tax savings 20%				0.80	0.64	0.51	0.41	0.33	0.26

Working 08: Calculating the Cost of equity									
Rf	9%								
Market Risk premium	3%								
Beta	1.2								
Additional premium for Local investments	3%								
Ke (For local investments)	**15.60%**								
Foreign investments	2%								
Ke (For foreign investments)	**17.60%**								

Working 09	Exchange Rate BDT/SAR							
BDT/SAR	29.333	31.048	32.830	34.735	36.799	38.978	41.203	43.580

Answer to the Question# 1(e):

The NPV for both projects is summarized below:
Investing in Mr. Pallab's business – BDT 20 million
Investing in Saudi Arabian operation – BDT 3,769 million
It shows that investing in Saudia Arabia operations has comparatively a higher NPV over the local investment. Hence, the same is recommended.
The following qualitative factors are important and should be considered:

- The accuracy of selling price (SP) and variable cost (VC) as the slightest change in their values will have a major impact on the final result.
- The exchange rates have been calculated based on the interest rate parity. However, there are many other factors that determine the exchange rate.
- It is also important to recognize that his father's business is also involved in household engineering items. Therefore, it is worthwhile to consider the application of the theory of "comparative advantage" in the context of manufacture of at least some basic components in Bangladesh rather than in KSA and thereby NPV of the KSA project could be improved.

Finally, the question of maximizing Reza's wealth also matters. Any project with + NPV would create wealth, but Reza should not blindly follow NPV as the decision criterion. Qualitative factors, and other underlying factors articulated in the Balanced Score Card and Porters 5 forces model, need due consideration in maximizing wealth, e.g., Entry barriers, industry attractiveness, etc… and if market believes that the project is not worthwhile, risks and costs will be higher. In essence, any project of this nature or magnitude requires recognition of features beyond the projects mere NPV.

Answer to the Question# 2(a) (i):

The number of new shares to be issued = 100 million (100 x 1/1)
The price per share = BDT 10 (1,000/1000)
This represents a discount on the current share price of 50% or BDT 10. (10/20)

Answer to the Question# 2(a) (ii):

The theoretical ex rights price is:

	Number of shares	**Value per share BDT**	**Number x value £**
Existing shares	1	20	20
New shares	1	10	10
Total shares	**2**	**Total value**	**30.00**
The theoretical ex rights price = BDT 15 (30/2)			

Answer to the Question# 2(a) (iii):

The actual share price will depend on the market's reaction to the rights issue, eg fully taken up, and whether the proceeds are invested in positive net present value projects. The net present value of the projects could be incorporated in the theoretical ex-rights price of BDT 15 giving a more realistic estimate of the actual share price post rights issue.

Answer to the Question# 2(b) (i):

The yield to maturity of Panama's debentures is:

105

The ex-interest price of Panama's debentures is

Years	Cash Flow	Factors @ 1%	PV BDT	Factors @ 6%	PV BDT
0	-105		-105		-105
01-05	7	4.853	33.971	4.212	29.484
5	100	0.951	95.1	0.747	74.7
			24.07		**-0.82**
IRR	**5.84%**				

The issue price of Panama's debentures will be:

The annuity factor for six years	4.943
The six year present value factor at 4.18%	0.712
The issue price =	110.74
The total nominal value of the debentures to be issued	902.98 million

Panama and Atlantic are in the same industry sector, so it is reasonable to assume that the yield to redemption of 5.84% is acceptable. However the financial risk of Atlantic might be different to Panama and this should be reflected in the yield to redemption.

Answer to the Question# 2(b) (ii):

Also Atlantic's debentures mature in five years and Panama's debentures mature in six years. It is likely that investors in Panama's debentures would require a higher yield to redemption than 5.84%, probably over 7% (5.84%/5 *6).

Answer to the Question# 2(c):

The gearing and interest cover ratios of Panama immediately after the debenture issue will be as follows:

Interest		72.24	
Interest cover		3.05	times

Gearing by market values assuming the current market price per share:

Market capitalization		2,000	million
Gearing		50%	
Current EPS			1.760
EPS with a rights issue			0.880
EPS with a debenture issues			1.182

In time both interest cover (more operating profits) and gearing (greater equity value) are likely to improve with the acceptance of positive NPV projects and any favourable market reaction to the issuance of debt and its tax shield (see below)

General advantages and disadvantages of debt v equity, include: Control issues; obligation to return capital; interest v dividends (including tax relief); issue costs; liquidation of the investment (can the investor get out

easily); risk/reward.

Answer to the Question# 2(d):

Analysis:

The company will have a gearing ratio of 50% and an interest cover of 3.05 times. Gearing is between the industry maximum and average of 55% and 45% respectively, but near to the maximum; interest cover is less than industry average and near to the minimum.

Since this is the first time that Panama has borrowed, both shareholders and the stock market might be concerned and prefer these ratios to be around the averages or better. Some shareholders might be attracted to investing in Panama because currently it has no gearing. However, if the BDT 1 billion is to be invested in positive NPV, projects both shareholders and the stock market should welcome the company borrowing.

Borrowing should reduce the current cost of capital of the company since debt is generally less expensive than equity because it is less risky than equity for the debt holders. Also the company receives tax relief on the interest that it pays. Because there is increased financial risk when a company borrows the shareholders may require a higher return but this is unlikely to offset the cheaper proportion of debt finance. The company value should increase as a result of the cost of capital reducing and new funds being invested in positive NPV projects.

Advice. It would be prudent for the company to restrict its borrowing to the industry average gearing level especially since its interest cover would be near to the minimum for the industry. I would advise the company not to borrow the full BDT 1 billion, perhaps this could be achieved by revising its plans for raising the finance. For example, an issue of both debt and equity to ensure that gearing and interest cover ratios are more favorable or selling surplus assets.

Answer to the Question# 3(a):

The apparent contradiction between the importance of dividends and the irrelevance of dividend policy stems from different perspectives and theories in finance. From an investor's point of view, dividends can be important as they provide a steady stream of income. However, from a company's perspective, the dividend policy may be considered irrelevant in certain financial theories, such as the Modigliani-Miller theorem. This theorem suggests that, under certain assumptions, the value of a firm is determined solely by its earning power and risk, and not by how it distributes profits (dividends or retained earnings). In practice, though, dividend policy can influence investor perceptions and behavior, even if it doesn't affect the firm's overall value.

Answer to the Question# 3(b):

The bird-in-the-hand argument implies that receiving a dividend today is safer than the uncertain prospect of a capital gain in the future. The fallacy in this argument lies in the assumption that dividends are inherently safer. In reality, the safety of dividends depends on the financial health of the company. A company paying high dividends may not necessarily be a safer investment if it compromises its ability to reinvest in the business or respond to changing market conditions. Capital gains, while uncertain, can also result from a company's growth and appreciation, making the bird-in-the-hand argument an oversimplification.

Answer to the Question# 3(c):

The fallacy in the argument that a firm can boost its share price by increasing its dividend payout ratio lies in assuming a direct cause-and-effect relationship between dividends and share price. While investors may prefer high dividends, artificially increasing the payout ratio may not necessarily lead to a sustained increase in share price. The long-term value of a company is influenced by various factors, including its ability to generate future earnings and manage risk. Relying solely on dividends as a strategy may neglect the importance of these other value drivers.

Answer to the Question# 3(d) (i):

Shofiq is demonstrating a preference for capital gains and stock price appreciation, while Sarah is demonstrating a preference for steady income in the form of dividends.

Answer to the Question# 3(d) (ii):

Shofiq could argue that Neotech's focus on reinvesting earnings for growth has led to a steady increase in its stock price, reflecting the company's potential for future capital gains. He could highlight Neotech's historical performance and growth prospects as reasons for Sarah to consider the stock.

Answer to the Question# 3(d) (iii):

Shofiq's argument might not convince Sarah if she has a strong preference for regular income in the form of dividends. If Sarah relies on dividends for her income needs and is risk-averse, the potential for capital gains may not align with her investment goals and preferences. Additionally, the absence of a dividend history in Neotech may be a concern for Sarah if she values a consistent income stream.

Answer to the Question# 4(a) (i):

First, we need to calculate the current sales and the change in sales. The current sales are next year's sales divided by one plus the growth rate, so:

Current sales = Next year's sales / (1 + g)
Current sales = Tk. 420,000,000 / (1 + .10)
Current sales = Tk. 381,818,182

And the change in sales is:

Change in sales = Tk. 420,000,000 - 381,818,182 = Tk. 38,181,818

We can now complete the current balance sheet. The current assets, fixed assets, and short-term debt are calculated as a percentage of current sales. The long-term debt and par value of stock are given. The plug variable is the additions to retained earnings. So:

Assets			Liabilities and equity	
Current assets	Tk.76,363,636		Short-term debt Long-term debt	Tk.57,272,727 Tk.120,000,000
Fixed assets	Tk.286,363,636		Liabilities and equity Common stock Accumulated retained earnings	Tk.48,000,000 137,454,545
			Total equity	Tk.185,454,545
Total assets	Tk.362,727,273		Total liabilities and equity	Tk.362,727,273

Answer to the Question# 4(a) (ii):

We can use the equation from the text to answer this question. The assets/sales and debt/sales are the percentages given in the problem, so:

$$\text{EFN} = \left(\frac{\text{Assets}}{\text{Sales}}\right) \times \Delta \text{ Sales} - \left(\frac{\text{Debt}}{\text{Sales}}\right) \times \Delta \text{ Sales} - (\text{PM} \times \text{Projected sales}) \times (1 - \text{d})$$

EFN = (.20 +.75) × Tk.38,181,818 -(.15 × Tk.38,181,818) – [(.09 × Tk.420,000,000) × (1 - .30)]
EFN = Tk. 4,085,455

Answer to the Question# 4(a) (iii):

The current assets, fixed assets, and short-term debt will all increase at the same percentage as sales. The long-term debt and common stock will remain constant. The accumulated retained earnings will increase by the addition to retained earnings for the year. We can calculate the addition to retained earnings for the year as:

Net income = Profit margin × Sales
Net income = .09 (Tk.420,000,000)
Net income = Tk.37,800,000

The addition to retained earnings for the year will be the net income times one minus the dividend payout ratio, which is:

Addition to retained earnings = Net income (1—d)
Addition to retained earnings – Tk.37,800,000 (1 - 0.30)
Addition to retained earnings = Tk.26,460,000

So, the new accumulated retained earnings will be:
Accumulated retained earnings = Tk.37,454,545 + 26,460,000
Accumulated retained earnings – Tk.163,914,545

The pro forma balance sheet will be:

Assets	Amount Tk		Liabilities and equity	Amount Tk
Current assets	84,000,000		Short-term debt	63,000,000
			Long-term debt	120,000,000
Fixed assets	315,000,000		Common stock	48,000,000
			Accumulated retained earnings.	163,914,545
			Total equity	Tk.211,914,545
Total assets	399,000,000		Total liabilities and equity	Tk.394,914,545

The EFN is:
EFN = Total assets - Total liabilities and equity
EFN = Tk. 399,000,000 - 394,914,545
EFN = Tk. 4,085,455

Answer to the Question# 4(b) (i):

The sustainable growth rate (SGR) can be calculated using the formula:

$$SGR = ROE \times (1 - \text{Payout Ratio})$$

Given that the ROE is 13.1 percent and the payout ratio is 40 percent, we can substitute these values into the formula:

$$SGR = 0.131 \times (1 - 0.4)$$

$$SGR = 0.131 \times 0.6$$

$$SGR = 0.0786 \text{ or } 7.86\%$$

Therefore, the company's sustainable growth rate is 7.86 percent.

Answer to the Question# 4(b) (ii):

Yes, the company's actual growth rate can be different from its sustainable growth rate. The sustainable growth rate represents the rate at which a company can grow its sales, earnings, and dividends without having to increase debt or equity. The actual growth rate, on the other hand, depends on various external factors such as market conditions, competition, and the company's strategic decisions. The actual growth rate may exceed the sustainable growth rate if the company takes on additional debt or equity to finance growth, or if external conditions are favorable. Conversely, it may be lower if the company faces challenges or opts for a more conservative growth strategy.

Answer to the Question# 4(b) (iii):

The company can increase its sustainable growth rate by taking actions that positively impact its return on equity (ROE) or by reducing its payout ratio. Here are some ways:

- **Improve Profitability:** Increasing the company's profitability, as measured by ROE, will contribute to a higher sustainable growth rate. This can be achieved by improving operational efficiency, increasing sales, or managing costs effectively.
- **Retain More Earnings:** Since the sustainable growth rate is influenced by the payout ratio, reducing the percentage of earnings paid out as dividends allows the company to retain more earnings for reinvestment. This can be achieved by either reducing dividend payments or by not increasing them at the same rate as earnings growth.
- **Debt Financing:** If the company can secure debt financing at favorable terms, it might consider using debt to fund expansion projects. This can boost the sustainable growth rate, but it also introduces financial risk.
- **Efficient Use of Assets:** If the company can improve its asset turnover (sales generated per unit of assets), it will contribute to higher ROE and, consequently, a higher sustainable growth rate.

It's essential for the company to strike a balance between growth and financial stability to ensure sustainable and healthy long-term development.

Answer to the Question# 5(a):

A company is borrowing therefore it will buy an FRA which locks it into a fixed borrowing rate. The spread by the bank will allow the bank to make a profit, thus:
- Depositors will receive the lower rate : 4.56%
- Borrowers will pay the higher rate : 6.59%

If the interest rates in December rise to 7.65% the bank will reimburse the company:
= (7.65% -6.59%) x 3/12 x Tk. 1 million = Tk. 2,650

Answer to the Question# 5(b) (i):

With SOFR at 12% for the next year, the cost with the swap would be as follows:

	Tk. '000
Interest: 10,000 x (14-13+12+1.5)%	1,450
Fees	10
	1,460

With SOFR at 12% for six months and at 10% thereafter, the cost with the swap would be as follows:

	Tk. '000
Interest: 10,000 x (14-13+(12+10)/2+1.5)%	1,350
Fees	10
	1,360

Without the swap, the cost would be Tk. 10m x 14% = Tk. 1.4m
The swap would therefore be disadvantageous to ZN Ltd if SOFR were to remain at 12% throughout the year but advantageous were SOFR to fall to 10% after six months.

Answer to the Question# 5(b) (ii):

Possible new terms would be for ZN Ltd to receive 13% and SOFR plus 1.5%. The net cost would be SOFR plus 0.5%+(14%-13%) = SOFR plus 1.5%, which is 0.5%less than the rate at which the company could raise floating rate debt.

MM Ltd would then effectively have fixed rate debt at 13% + (1-0.5)% = 13.5%, which is 0.5% less than the rate at which it could otherwise have such debt.

Answer to the Question# 6(a):

Spot rate	234.46	-	234.78
3 months premium	2.72	-	2.56
	231.74	-	232.22

The company is selling yen in three months' time, and so must use the dealer's buying rate of 232.22
Therefore, amount receivable = 4,680m / 232.22 = Tk. 20,153,303

Answer to the Question# 6(b) (i):

	31-Dec-2022		
Exchange rate	Tk. 92/US$	Tk. 105/US$	Tk. 85/US$
	Tk. '000	Tk. '000	Tk. '000
Overseas assets in Tk. value of US$ 1 million	92,000	105,000	85,000
Bangladeshi assets	60,000	60,000	60,000
	152,000	165,000	145,000
Bangladeshi loan	(110,000)	(110,000)	(110,000)
	42,000	55,000	35,000
Gearing ratio	110/152	110/165	110/145
	72.4%	66.7%	75.9%

Answer to the Question# 6(b) (ii):

The increase in gearing caused by the drop in the Taka value of net assets or equity could be mitigated by a similar drop in the Taka value of the debt. Thus, financing the overseas assets raising capital in US$ would stop the relative gearing level rising by as such.

---The End---